IndicCorp v2 corpus provides large monolingual datasets for different Indian languages. The corpus can be accessed from the link below:
https://huggingface.co/datasets/ai4bharat/IndicCorpV2

You need to design a language identification model that identifies the language of a given text. For that you have to construct a dataset for language identification using the above repository. IndicCorpV2 provides texts for 23 languages along with the written script (some languages are written in different scripts). Take 1000 sentences from each language. You also need to take care about the tokenization. Each sample in the corpus is not a sentence, it is usually a paragraph. So tokenize the paragraph into sentences and create your data. Split the data into 3 parts: train, validation, and test in a ratio of 80: 10: 10. Follow stratified sampling so that you have same number of training, validation, and test samples for each language. Use the same labels as given in the repository.

1. Represent each sentence using TF-IDF vectors.
    a. Your TF-IDF vectors are composed of word and character n-grams.
        i. Word Unigrams and Word Bigrams
        ii. Character 2-grams, 3-grams, and 4-grams
    b. You cannot use any predefined classes under sklearn. Build you own TF-IDF vectorizer.
2. Design a logistic regression based language identifier using the above mentioned TF-IDF vectors.
    a. You cannot use any predefined classes under sklearn. Design your own logistic regression classifier.
3. Evaluate your model using macro-F1 evaluation metric. You cannot use any predefined classes under sklearn for the evaluation.